Exercise Set 1

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The choice of group matters. In the lecture, the discrete logarithm problem (DLP) was considered in an arbitrary cyclic group $\mathbb{G}$ of prime order q generated by an element g; we write $\mathbb{G} = \langle g \rangle$ for the group generated by g. There, the best generic algorithms need $\mathcal{O}(\sqrt{q})$ group operations. In this exercise set we look at concrete groups in which the DLP is considerably easier, because the group has additional structure that a generic algorithm does not use. The main tools are pairings, which are also the basis of several constructions in modern cryptography. Parts marked $\star$ are intended as take-home exercises.

Bilinear pairings: Let $\mathbb{G}_1 = \langle P \rangle$, $\mathbb{G}_2 = \langle Q \rangle$ and $\mathbb{G}_T$ be cyclic groups of the same prime order q, the first two written additively and $\mathbb{G}_T$ multiplicatively. A *(bilinear) pairing* is a map

$$e : \mathbb{G}_1 \times \mathbb{G}_2 \longrightarrow \mathbb{G}_T$$

with the following three properties.

Bilinearity $e([a]P', [b]Q') = e(P', Q')^{ab}$ for all $P' \in \mathbb{G}_1$, $Q' \in \mathbb{G}_2$ and $a, b \in \mathbb{Z}$.
Non-degeneracy $e(P, Q) \neq 1$.
Efficiency e can be evaluated in polynomial time.

If $\mathbb{G}_1 = \mathbb{G}_2$ the pairing is called *symmetric*, and we then write $\mathbb{G}$ for $\mathbb{G}_1 = \mathbb{G}_2$.

For Problems 1 and 2 no knowledge of how a pairing is constructed is needed. Problem 3 uses a concrete pairing, the Weil pairing, which is introduced there.

Problem 1. (Generic hardness vs. hardness)

(a) Consider the group $(\mathbb{Z}_N, +)$. Fix $g \in \mathbb{Z}_N$ with $\gcd(g, N) = 1$ and let $h = x \cdot g \bmod N$. Give an algorithm that recovers x and state its cost.

(b) Every cyclic group of order N is isomorphic to $(\mathbb{Z}_N, +)$. Why does part (a) not break the DLP in *every* cyclic group of order N?

The remaining problems show elliptic curves whose representation can be exploited in a less obvious way using for example pairings.

Problem 2. (Pairings as a black box)

(a) Let $e : \mathbb{G} \times \mathbb{G} \to \mathbb{G}_T$ be a symmetric pairing as defined above, where $\mathbb{G} = \langle P \rangle$. On input $(P, [a]P, [b]P, [c]P)$, give an efficient test for the DDH problem, i.e. whether $c \equiv ab \pmod{q}$.

(b) Now let $e : \mathbb{G}_1 \times \mathbb{G}_2 \to \mathbb{G}_T$ be a general, not necessarily symmetric, pairing as defined above, with $\mathbb{G}_1 = \langle P \rangle$ and $\mathbb{G}_2 = \langle Q \rangle$. Let $R = [x]P \in \mathbb{G}_1$ for unknown x. Construct in polynomial time an instance of the DLP in $\mathbb{G}_T$ whose solution is x. What does this say about the hardness of the DLP in $\mathbb{G}_1$ versus $\mathbb{G}_T$?

(c) $\star$ Given a group $\mathbb{G} = \langle P \rangle$ with a symmetric pairing, design a one-round three-party key exchange. That is, how do Alice, Bob and Charlie holding secrets a, b, c agree on a shared secret key if each can only broadcast *one* message?

(d) $\star$ Let $H : \{0,1\}^* \to \mathbb{G}_1$ be a hash function onto $\mathbb{G}_1$. The secret key is x, the public key $\mathrm{pk} = [x]Q \in \mathbb{G}_2$, and the signature on m is $\sigma = [x]H(m) \in \mathbb{G}_1$. Give the verification equation and prove correctness.

(e) In all pairings we know how to build, $\mathbb{G}_1$ is a subgroup of an elliptic curve group $E(\mathbb{F}_p)$ and $\mathbb{G}_T$ is a subgroup of $\mathbb{F}_{p^k}^*$ for some $k \geq 1$. The best known algorithm for the DLP

in $\mathbb{G}_1$ is Pollard's rho method with $\mathcal{O}(\sqrt{q})$ group operations, while the DLP in $\mathbb{F}_{p^k}^*$ can be solved with the Number Field Sieve in subexponential time $L_{p^k}(1/3)$. Here $\mathbb{F}_{p^k}^*$ is the multiplicative group of the finite field with p^k elements, k is called the *embedding degree* (defined before Problem 3), and $L_{p^k}(1/3) = \exp\left(c\,(\ln p^k)^{1/3}(\ln\ln p^k)^{2/3}\right)$ for a constant c, a function that grows slower than any exponential in $\log p^k$. Given part (b), what does this mean for the choice of parameters if we *want* to use a pairing?

Background for Problem 3. Let E be an elliptic curve over $\mathbb{F}_p$ and $q \neq p$ a prime. $\#E(\mathbb{F}_p)$ denotes the number of points of E with coordinates in $\mathbb{F}_p$ (including $\mathcal{O}$), and $\overline{\mathbb{F}}_p$ the algebraic closure of $\mathbb{F}_p$, i.e. the union of all finite extensions $\mathbb{F}_{p^j}$; a point of $E(\overline{\mathbb{F}}_p)$ has coordinates in some $\mathbb{F}_{p^j}$. The following facts may be used without proof.

Counting	$\#E(\mathbb{F}_p)$ can be computed in polynomial time (e.g. SEA algorithm), and $\#E(\mathbb{F}_p)$ determines $\#E(\mathbb{F}_{p^k})$ for every k by a closed formula. Random points of $E(\mathbb{F}_{p^k})$ can be sampled efficiently.
Extensions	$E(\mathbb{F}_p) \subseteq E(\mathbb{F}_{p^k})$ for every $k \geq 1$, with the same addition formulas. $\mathbb{F}_{p^k}^*$ is cyclic of order $p^k - 1$; a cyclic group of order M has a subgroup of order q iff $q \mid M$.
Torsion	The q-torsion $E[q] := \{S \in E(\overline{\mathbb{F}}_p) : [q]S = \mathcal{O}\}$ is a group isomorphic to $\mathbb{Z}_q \times \mathbb{Z}_q$ with q^2 points; a point $P \in E(\mathbb{F}_p)$ of order q generates only one of its $q+1$ subgroups of order q. The *embedding degree* k is the smallest integer with $q \mid p^k - 1$. If $q \nmid p - 1$, then $E[q] \subseteq E(\mathbb{F}_{p^k})$.
Weil pairing	There is a map $e_q : E[q] \times E[q] \to \mu_q := \{\zeta \in \overline{\mathbb{F}}_p : \zeta^q = 1\}$ that is bilinear, *alternating* ($e_q(S,S) = 1$ for all S), non-degenerate, and computable by Miller's algorithm in time polynomial in $\log p^k$.

Problem 3. (The MOV attack)

We now apply Problem 2(b) to build an attack on the DLP for certain elliptic curves. This attack is due to Menezes, Okamoto and Vanstone (1991).

Let $P \in E(\mathbb{F}_p)$ be a point of prime order $q \neq p$ with $q \nmid p - 1$, let $R = [x]P$, and let k be the curve's embedding degree. The pairing will be e_q with $\mathbb{G}_1 = \langle P \rangle$ and $\mathbb{G}_T = \mu_q$.

(a) Show that $\mu_q \subseteq \mathbb{F}_{p^k}^*$ and that k is minimal with this property. Conclude that all points of $E[q]$ and all pairing values live in $\mathbb{F}_{p^k}$.

(b) Fix $T \in E[q]$ and define $\varphi_T : \langle P \rangle \to \mu_q$, $S \mapsto e_q(S,T)$. Show that φ_T is a group homomorphism, and determine exactly when it is injective.

(c) By part (b), we want a point $T \in E[q]$ with $e_q(P,T) \neq 1$, so that φ_T is injective. Show that every $T \in \langle P \rangle$ is a bad choice. Then compute the probability that a uniformly random $T \in E[q]$ is bad.

(d) Let $N := \#E(\mathbb{F}_{p^k})$. Explain why $q^2 \mid N$. Assume $q^3 \nmid N$ and let $T' \in E(\mathbb{F}_{p^k})$ be uniformly random. Show that $T := [N/q^2]\,T'$ is uniformly random in $E[q]$. Why does $[N/q]\,T'$ not work?

(e) Assemble (a)–(d) into an algorithm that, on input (E, P, R), reduces the ECDLP in $\langle P \rangle$ to a DLP in $\mathbb{F}_{p^k}^*$ and then solves the latter with the Number Field Sieve. State the cost of each step. For $\log_2 p = 256$ and $q \approx p$, for which k is this faster than Pollard rho on the curve? You may use that the NFS in a finite field of about 3072 bits costs roughly 2^{128} operations.